DocuMind AI Architecture & Retrieval Specification

Document ID: SPEC-2025-V1 | Classification: Internal Architecture Guide

SECTION 1: RETRIEVAL-AUGMENTED GENERATION (RAG) PIPELINE

DocuMind is an enterprise knowledge assistant designed to extract information from unstructured documents without hallucination. When a user submits an inquiry, the query vector is matched against document vectors using cosine similarity.

Key parameters for chunking:
- Chunk Size: 600 characters per segment
- Overlap: 120 characters between consecutive segments
- Boundary Detection: Splits on paragraph, sentence, and phrase endings.

The vector representation is computed using Google Gemini text-embedding-004, which outputs 768-dimensional normalized dense vectors.

SECTION 2: VECTOR DATABASE & STRICT GROUNDING

For production deployments on Vercel, vectors are persisted in Upstash Vector,
a serverless HTTP REST-compatible vector database.
For local developer workstations, a local vector store runs seamlessly with zero setup.

SECTION 3: GROUNDING & REFUSAL POLICY

The generative model (gemini-1.5-flash) is configured with low temperature (0.1).
If the retrieved context does not contain the answer, the assistant MUST respond:
"I couldn't find this information in your uploaded documents."

SECTION 4: COMPLIANCE & RETENTION

- Data Retention Window: 90 days for audit logs.
- Designated Data Privacy Officer: Dr. Sarah Lin (privacy@documind.example).
- Security Audits: Conducted bi-annually by third-party auditors.